Name: _____

Section: _____

Instructions: Show supporting work to receive full credit for each problem. No calculators, notes, or other external resources are allowed. The quiz will be graded out of 10 points.

1. Solve the given linear ODE for the unknown function $V(t)$. Remember to include the domain of existence of the solution.

$$\begin{cases} \frac{dV}{dt} - 2tV = V + 4t + 2, \\ V(0) = 3 \end{cases} .$$